PRD: Lyrics Audio Player for Android and Android Auto

*Working title: **TurqPlay** · Version 1.0 draft · Platform: Android phone, tablet, Android Auto*

1. Overview

TurqPlay is a simple, modern local audio player for Android that shows time-synced lyrics while music plays. It supports importing lyric files, manually syncing lyrics, and automatically generating lyrics from audio. It integrates with Android Auto so the player appears on the car display when the phone is connected, with safe, driver-friendly controls.

Vision: the cleanest way to listen to your own music library and sing along, at home or in the car.

2. Goals and Non-Goals

Goals

- Play local audio files with a clean, turquoise-themed UI.
- Display synced lyrics on the phone and tablet; show a safe lyric experience in the car.
- Let users import, edit, sync, or auto-generate lyrics.
- Appear automatically in Android Auto as a media app.
- Provide Equalizer, Bass, and Treble controls and theme selection.

Non-Goals (v1)

- Music streaming service or music store.
- Social features, playlists sharing, podcasts.
- Video playback.
- iOS or CarPlay.

3. Target Users

Persona	Need
Daily commuter	Plays own MP3/FLAC library in the car, wants easy, safe controls

Persona	Need
Language learner / singer	Wants accurate synced lyrics to follow along
Audio enthusiast	Wants EQ, bass and treble tuning on a lightweight app

4. Key Platform Constraint: Android Auto (Read First)

These points shape the whole design and should be confirmed with the team early.

1. **Bluetooth alone does not launch Android Auto.** Android Auto starts over a USB cable, or over **wireless Android Auto** (Bluetooth for pairing plus Wi-Fi for projection) on supported head units and phones. A plain Bluetooth audio connection (A2DP) plays sound, but shows no Android Auto UI. The app should still work in that case through standard media session metadata (title, artist, album art, track controls) on the car stereo.
2. **Media apps cannot draw custom UI on the car screen.** Android Auto renders its own templates driven by a media session. The app supplies metadata, browse tree, playback state and a limited set of custom actions. Full-screen custom lyrics, sliders, and EQ panels are not allowed on the car display.
3. **Driver distraction rules apply.** Scrolling or animated text, and dense text, are restricted while driving. Lyrics in the car must follow Google's Android Auto media app guidelines.
4. **Consequence for lyrics:** full synced lyrics live on the phone/tablet UI (and are available in the car only in the compliant forms listed in section 7.6). Equalizer, Bass and Treble are adjusted on the phone, and their settings apply to car playback as well.

5. Product Scope Summary

Area	Phone / Tablet	Android Auto
Play / Pause, Next, Back	Yes	Yes
Seek slider	Yes	Yes (progress bar from media session)
Synced lyrics	Full screen, auto-scroll	Limited, compliant form (see 7.6)
Equalizer / Bass / Treble	Full controls	Applies saved settings; optional preset custom action

Area	Phone / Tablet	Android Auto
Theme selection	Yes	Car UI follows Android Auto theme; app accent via artwork only
Lyrics sync / generate	Yes	No (disabled while connected to the car)
Library browse	Yes	Simple browse tree (Songs, Albums, Artists, Playlists)

6. User Stories

- As a listener, I want to play a song and see lyrics highlight line by line so I can sing along.
- As a user, I want to import an .lrc file for a song so lyrics sync to the audio.
- As a user with plain-text lyrics, I want to tap through the song to timestamp each line.
- As a user with no lyrics, I want the app to generate them automatically.
- As a driver, I want to connect my phone to my car and find the player in Android Auto without setup.
- As a driver, I want to browse and control playback with big, simple, safe controls.
- As a user, I want to adjust bass, treble and equalizer presets to my taste.
- As a user, I want to choose light, dark or system theme with a turquoise look.

7. Functional Requirements

7.1 Audio Playback

- Supported formats: MP3, AAC/M4A, FLAC, OGG, WAV, OPUS (via Media3/ExoPlayer).
- Controls: Play/Pause, Previous, Next, seek slider, shuffle, repeat (off / all / one).
- Gapless playback, background playback, lock-screen and notification controls.
- Audio focus handling: pause on calls, duck for navigation prompts, resume afterwards.
- Resume last position and queue after app restart or Android Auto reconnect.
- Handle headset unplug and Bluetooth disconnect by pausing.

7.2 Library

- Scan device storage via MediaStore; tabs for Songs, Albums, Artists, Folders, Playlists.
- Search by title, artist or album.

- Read ID3/Vorbis tags and embedded artwork; embedded lyrics tags (USLT / SYLT) are used when present.

7.3 Now Playing Screen (Phone)

- Album art, title, artist.
- Large Play/Pause button, Previous and Next buttons.
- Seek slider with elapsed and remaining time.
- Lyrics button toggles the full-screen lyrics view; swipe up on the card also opens it.
- Quick access to Equalizer and Theme from the toolbar.

7.4 Lyrics Display

- Synced mode: current line highlighted in turquoise, auto-scrolls, previous and upcoming lines dimmed.
- Unsynced mode: plain scrollable text when no timestamps exist.
- Tap a line to seek to that timestamp.
- Adjustable font size; screen stays awake while lyrics are open (optional setting).
- Offset control (plus or minus milliseconds) to fix global drift.
- Supported formats: **LRC** (line sync), **Enhanced LRC** (word sync, v1.1), **TXT** (unsynced), **SRT** (import as lyrics).

7.5 Lyrics Sources and Sync (Provision to Sync or Generate)

Priority order when opening a song:

1. Embedded lyrics in the file tags.
2. Sidecar file with the same name in the same folder (song.lrc).
3. Previously saved or generated lyrics in the app database.
4. Optional online lookup (user opt-in) from a licensed lyrics provider or open LRC database, matched by title, artist, album and duration.
5. Otherwise show "Add lyrics" with the options below.

A. Import lyric file

- Pick .lrc, .txt or .srt via the system file picker; attach to the current song.
- Validate encoding (UTF-8, UTF-16, common legacy encodings) and show a preview before saving.

B. Manual sync editor

- User pastes or imports plain text lyrics.

- Playback runs while the user taps "Mark line" at each line start; the app records timestamps.
- Fine-tune: nudge any line by 100 ms steps, replay a line, undo, save as .lrc.

C. Automatic lyric generation

- Pipeline: speech recognition on the audio track, then segment into lines with timestamps, saved as LRC.
- Two modes:
  - **On-device** (private, works offline): a compact speech model such as a Whisper-class model, running in the background with progress notification. Slower, best on newer devices.
  - **Cloud** (opt-in): faster and more accurate for music; user is told audio is uploaded and must consent.
- Alignment option: if the user supplies plain-text lyrics, perform forced alignment of the text to the audio for accurate timestamps (often more accurate than pure transcription for sung vocals).
- Set expectations in the UI: accuracy varies with music, language and vocals. Generated lyrics are labeled "Auto-generated" and are editable.
- Language selection or auto-detect; v1 languages: English plus a configurable list.
- Batch generation for a selected album or folder while charging and on Wi-Fi.
- Generation is available only on the phone and tablet, never in Android Auto.

D. Storage

- Lyrics are saved to a local database keyed by file hash, with an option to also write a sidecar .lrc next to the file.
- Export and share .lrc.

7.6 Android Auto Integration

- Implemented as a media app using a Media3 `MediaLibraryService` with a `MediaSession`, declared for Android Auto in the manifest.
- Appears in the Android Auto launcher automatically when the phone connects by USB or wireless Android Auto; no extra pairing steps inside the app.
- Browse tree: Recently Played, Songs, Albums, Artists, Playlists, with limited item counts per level and paging, per Google's guidelines.
- Voice commands via Google Assistant: "Play [song/artist] on TurqPlay" through `onPlayFromSearch`.
- Now Playing on the car screen: title, artist, album art, Play/Pause, Previous, Next, progress bar, shuffle and repeat custom actions.

- **Lyrics in the car (compliance-first):**
  - Default: show the current lyric line in a metadata field supplied to the media session (for example the subtitle), updated per line, rather than animated scrolling. This must be validated against the current Android Auto quality guidelines and Google Play review before launch, because policies on lyrics text in metadata may restrict it.
  - Setting: "Show lyrics in car" (on or off), plus an automatic hide when the car reports driving state if required by the guidelines.
  - Fallback if disallowed: lyrics remain phone-only, with the phone showing the lyrics screen when parked.
- Bluetooth-only (no Android Auto): standard AVRCP metadata and controls work; optionally supply the current lyric line as the track title or artist metadata only if the user enables a clearly labeled setting, since many car stereos show those fields.
- Equalizer, Bass and Treble apply to the audio session regardless of where the controls were set.
- Lyrics sync and generation actions are blocked in the car; the app tells the driver to use the phone when parked.

7.7 Audio Effects: Equalizer, Bass, Treble

- **Equalizer:** 5 or more bands (device-dependent via `android.media.audiofx.Equalizer`), presets (Flat, Pop, Rock, Jazz, Classical, Vocal, Bass Boost, Custom), save custom presets.
- **Bass:** slider 0 to 100 percent using `BassBoost` or low-shelf equalization where unsupported.
- **Treble:** slider -10 to +10 dB, implemented as a high-shelf adjustment on the highest equalizer bands.
- On/off master toggle; reset to flat.
- Global settings with optional per-output profiles (speaker, headphones, Bluetooth, car).
- Fallback: if the device does not support an effect, hide or disable it with an explanation.
- Prevent clipping: apply a small automatic pre-gain reduction when boosting.

7.8 Theme and Appearance

- **Default theme: Turquoise.** Material 3 design, large touch targets, rounded shapes, generous spacing.
- Theme options: Light, Dark, System, and AMOLED black.
- Accent options: Turquoise (default), plus optional alternate accents later.

- Dynamic album-art tint on the Now Playing screen is optional and off by default so the turquoise identity remains.

Turquoise palette (proposal)

Token	Light	Dark
Primary	#00A99D	#40E0D0
On Primary	#FFFFFF	#00201D
Primary Container	#B2F0E8	#00504A
Secondary	#4A6360	#B1CCC8
Background	#F6FBFA	#0E1514
Surface	#FFFFFF	#151D1C
Lyric highlight	#00A99D	#40E0D0
Lyric dimmed	50% on-surface	50% on-surface

Check contrast to meet WCAG AA (4.5:1 for text).

8. UI Specification

Principles: simple, modern, one primary action per screen, minimal chrome.

Screens

1. **Home / Library:** bottom navigation (Library, Now Playing, Settings), mini-player above the nav bar.
2. **Now Playing:** artwork, title, artist, seek slider, Back / Play-Pause / Next, shuffle and repeat, buttons for Lyrics and Equalizer.
3. **Lyrics:** full-screen, large text, current line highlighted, compact mini controls at the bottom, overflow menu: Import lyrics, Sync lyrics, Generate lyrics, Adjust offset, Font size.
4. **Sync Editor:** lyric lines list, big "Mark line" button, playback controls, undo.
5. **Equalizer:** band sliders, preset chips, Bass and Treble sliders, on/off switch.
6. **Settings:** theme, lyrics sources, auto-generation mode, Android Auto options, storage, about.

Tablet: two-pane layout, library and queue on the left, Now Playing and lyrics on the right; landscape supported.

Car: rendered by Android Auto templates; the app supplies icons and artwork only. Custom icons must be monochrome and meet size guidelines.

9. Non-Functional Requirements

Category	Requirement
Performance	Cold start under 2 s on mid-range device; playback start under 500 ms; lyric highlight latency under 50 ms
Battery	Background playback under 3% per hour on mid-range device; generation tasks run only on charger by default
Size	APK/AAB under 25 MB excluding downloadable speech models
Compatibility	Android 8.0 (API 26) and above; phones, tablets, foldables
Reliability	Crash-free sessions above 99.5%; session recovers after Android Auto reconnect
Privacy	Local-first; no audio leaves the device without explicit consent; no ads or tracking
Accessibility	TalkBack labels, minimum 48 dp touch targets, scalable text
Localization	RTL support, Unicode lyrics, UI strings externalized

10. Technical Approach

- **Language and UI:** Kotlin, Jetpack Compose with Material 3.
- **Playback:** Jetpack Media3 (ExoPlayer, `MediaLibraryService`, `MediaSession`).
- **Android Auto:** media app declaration (`automotive_app_desc.xml`), Desktop Head Unit (DHU) for testing.
- **Effects:** `Equalizer`, `BassBoost`, `Virtualizer` bound to the ExoPlayer audio session ID.
- **Data:** Room database for library cache, lyrics and presets; DataStore for settings.
- **Lyrics parsing:** custom LRC parser (multi-timestamp lines, offset tag, enhanced word tags).

- **Auto-generation:** background WorkManager jobs; on-device ASR through ONNX Runtime or TensorFlow Lite; optional cloud endpoint.
- **Architecture:** MVVM, Hilt for dependency injection, Kotlin coroutines and Flow.
- **Testing:** unit tests for the LRC parser and sync logic, instrumented tests for playback, DHU manual test pass.

11. Release Plan

Phase	Scope
MVP	Local playback, library, Now Playing UI, turquoise theme with light/dark, LRC import and synced display, Android Auto media browsing and controls, Equalizer, Bass and Treble
v1.1	Manual sync editor, embedded and sidecar lyric detection, online lyric lookup (opt-in), car lyric line display (policy-approved)
v1.2	On-device auto-generation, forced alignment, batch jobs
v1.3	Cloud generation option, word-level lyrics, per-output EQ profiles, tablet polish

12. Success Metrics

- Percentage of played songs that show lyrics (target above 60% after 30 days).
- Android Auto session crash-free rate above 99.5%.
- Day-30 retention above 25%.
- Auto-generated lyrics edit rate and user rating of accuracy (target 3.5 out of 5 or better).
- Play Store rating of 4.3 or higher.

13. Risks and Open Questions

Risk / Question	Mitigation
Android Auto may not allow lyrics display beyond metadata	Validate early with DHU and Google's guidelines; keep phone-first lyrics as the core value
Automatic lyric accuracy on sung audio is lower than on speech	Use forced alignment when text is available; label results; make editing easy
Lyrics are copyrighted; online lookup and display carry licensing obligations	Use licensed or openly licensed providers; do not bundle lyrics; user-supplied files stay local
On-device models are large and slow on low-end devices	Optional download, device capability check, cloud fallback with consent
Users expect Bluetooth connection to open Android Auto	Onboarding explains USB or wireless Android Auto requirements
Equalizer support differs across devices and OEMs	Capability detection and graceful fallback
Play Store review for Android Auto media apps	Follow quality guidelines, prepare test instructions and demo video

Open questions

1. Which languages must auto-generation support at launch?
2. Is an online lyrics provider in scope for v1, and who owns the licensing?
3. Will the app be free, ad-free paid, or freemium (for example, premium for auto-generation)?
4. Should word-level (karaoke) highlighting be pulled into the MVP?

14. Acceptance Criteria (MVP)

- Connecting the phone via USB shows the app in Android Auto and playback, browse, Next and Back work.
- Play/Pause, Next, Back and the slider work on the phone, notification, lock screen and car.
- Importing an .lrc file displays highlighted lines within 50 ms of the audio position.
- Bass, Treble and Equalizer changes are audible immediately and persist across restarts.

- The turquoise theme is applied across all screens in light and dark modes with AA contrast.
- Audio pauses on calls and resumes correctly; no crashes through a 2-hour playback soak test.